Assessing neurologic system

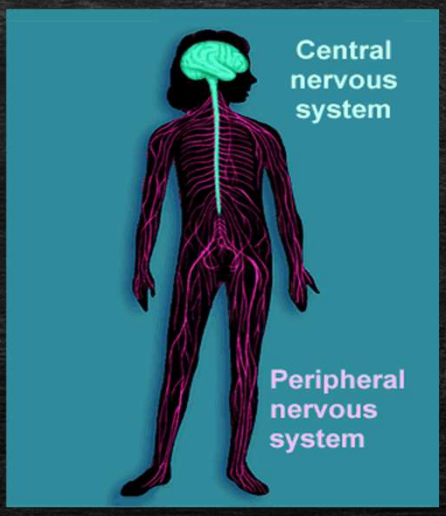
Weber: Chapter 25
NURS3120: Health Assessment and Promotion

Objectives

- Describe the structure and function of the central and peripheral nervous system
- Discuss risk factors associated with a cerebrovascular accident (CVA), and ways to reduce one's risks
- Differentiate between normal and anormal findings of the neurologic system
- Analyze the data from the interview and physical assessment of the neurologic system to formulate valid clinical judgments

Central nervous system: The brain & spinal cord

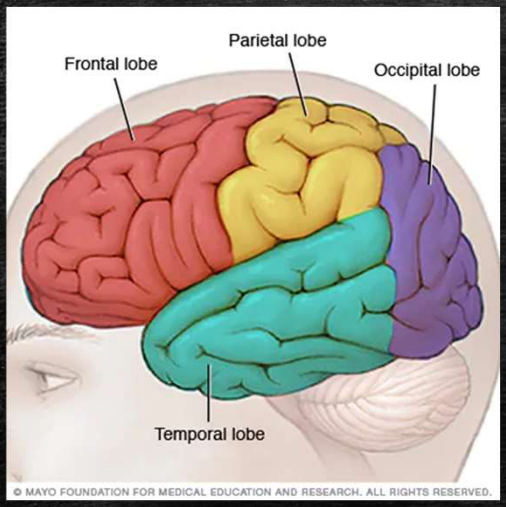
- The brain
  - 4 regions:
    - Cerebrum
    - Diencephalon
    - Brainstem
    - Cerebellum



The neurologic system is very complex and is responsible for coordinating and regulating all the body functions. It is composed of two structural components: the central nervous system and the peripheral nervous system. The CNS encompasses the brain and the spinal cord which are covered by meninges-three layers of connectives tissues that protect and nourish the CNS. The brain is composed of 4 regions:
Cerebrum
Diencephalon
Brainstem
Cerebellum

8 cervical, 12 thoracic, 5 lumbar, 5 sacral and 1 coccygeal nerve which is a total of 31 pairs of spinal nerves

Central nervous system



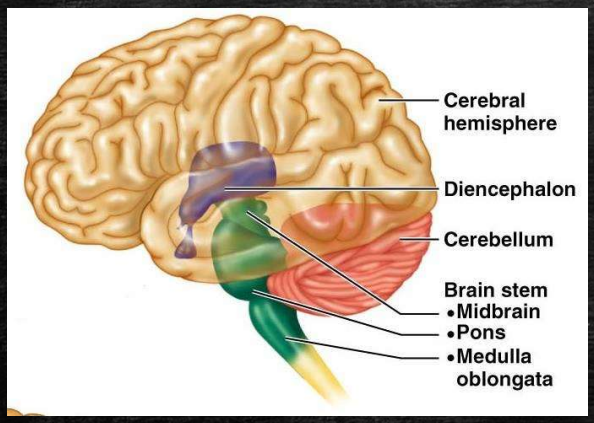
- Cerebrum
  - Divided into the left & right hemisphere
  - Each divided into 4 lobes
    - Frontal
    - Parietal
    - Occipital
    - Temporal
  - Wernicke's area
  - Broca's area

Cerebrum: is divided into the right and left cerebral hemispheres. Each hemisphere sends and receives impulses from the opposite sides of the body and consists of four lobes (frontal, parietal, temporal, and occipital) The lobes are composed of a substance known as gray matter, which mediates higher-level functions such as memory, perception, communication and initiation of voluntary movements. The aggregations of neuronal cell bodies, gray matter rims the surfaces of the cerebral hemispheres, forming the cerebral cortex.

Frontal: directs voluntary, skeletal actions (left side of lobe controls right side of body, and right side of lobe controls left side of body). It also influences communication (talking and writing) emotions, intellect, reasoning ability, judgment, and behavior. It contains the Broca's area which is responsible for speech.
Parietal: Interprets tactile sensations, including touch, pain temperature, shapes and two-point discrimination
Occipital: Influences the ability to read with understanding and is the primary visual receptor center
Temporal: Receives and interprets impulses from the ear. Contains Wernicke area which is responsible for interpreting auditory stimuli.

Why is this important to know?

Brain & spinal cord



- Basal ganglia
- Diencephalon
  - Thalamus
  - Hypothalamus
- Cerebellum
- Brainstem
- Spinal cord

Basal ganglia: a group of structures near the center of your brain that form important connections. These connections allow different areas of your brain to work together. The basal ganglia manage the signals your brain sends that help you move your muscles.

Diencephalon: lies beneath the cerebral hemispheres and consists of the thalamus and hypothalamus. Most sensory impulses travel through the gray matter of the thalamus which is responsible for screening and directing the impulses to specific areas in the cerebral cortex.

The hypothalamus is responsible for regulating body functions such as water balance, appetite, VS, sleep cycles, pain perception, and emotional status.

Cerebellum: Is located behind the brain stem and under the cerebrum. Its primary functions include coordination and smoothing of voluntary movements, maintenance of equilibrium, and maintenance of muscle tone.

Brainstem: located between the cerebral cortex and the spinal cord, the brain stem consists mostly nerve fibers and has three parts: the midbrain, pons, and medulla oblongata. The medulla oblongata contains the nuclei for cranial nerves, and has centers that control and regulate respiratory function, heart rate and force and blood pressure.

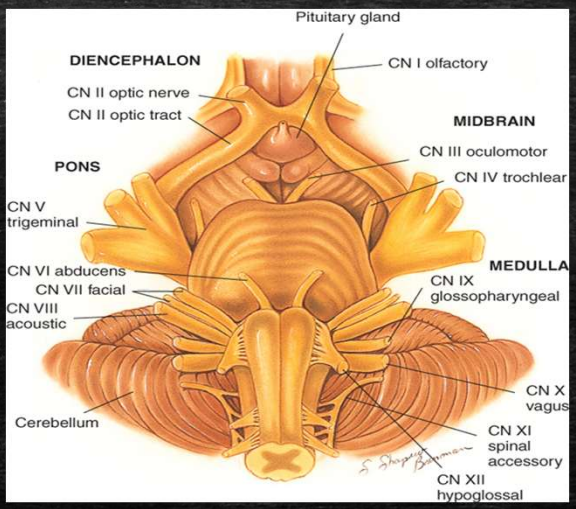
The **spinal cord:**

Extends from the medulla oblongata to L1/2 vertebrae

Conducts sensory impulses up ascending tracts to the brain, conducts motor impulses down descending tracts to neurons that stimulate glands and muscles throughout the

body, and is responsible for simple reflex activity.
8 cervical, 12 thoracic, 5 lumbar, 5 sacral and 1 coccygeal nerve which is a total of 31 pairs of spinal nerves.

Peripheral nervous system: Cranial nerves

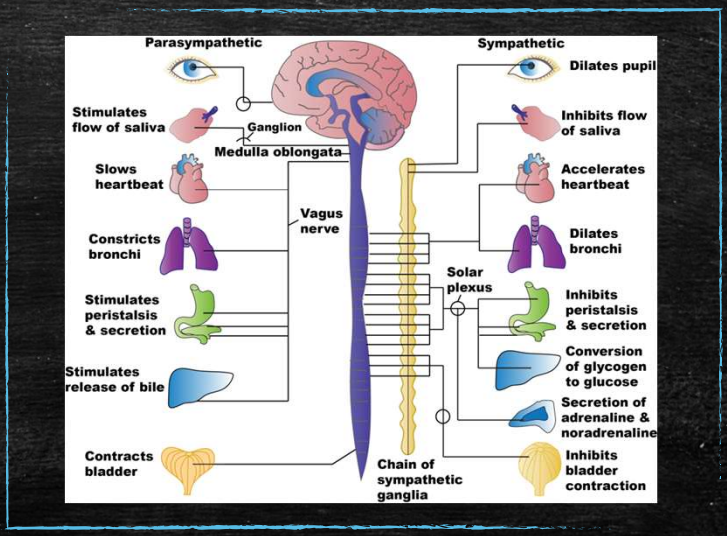


Carrying information to and from the CNS, the peripheral nervous system consists of 12 pairs of cranial nerves and 31 pairs of spinal nerves.

Twelve pairs of cranial nerves evolve from the brain or brain stem and transmit motor or sensory messages.

Cranial nerves also govern specialized functions like smell, vision, and hearing.

Autonomic nervous system



Autonomic nervous system impulses are carried by both cranial and spinal nerves. These impulses are carried from the CNS to the involuntary, smooth muscles that make up the walls of the heart and glands. The autonomic nervous system maintains internal homeostasis of the body, incorporates the sympathetic and parasympathetic nervous system. Sympathetic is the fight or flight and is activated during stress and elicits responses such as decreased gastric secretions, bronchiole dilation, increased pulse rate and pupil dilation. The parasympathetic nervous system (rest & digest) functions to restore and maintain normal body functions, for example by decreasing heart rate.

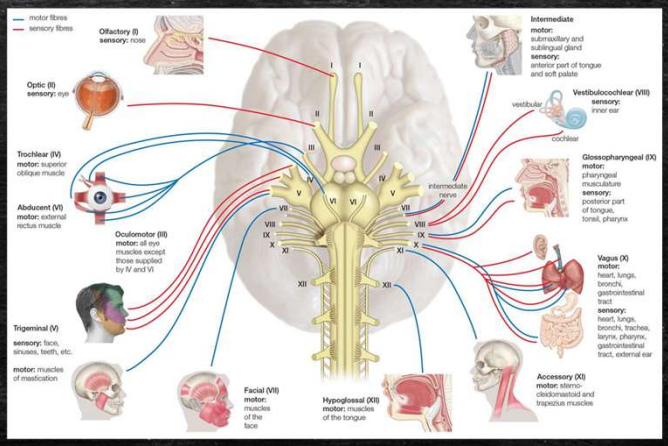
Collecting subjective data: Health history

- Headache
- Head injury
- Dizziness/Vertigo
- Seizures
- Tremors
- Weakness
- Incoordination
- Numbness/tingling
- Difficulty swallowing
- Difficulty speaking
- Significant past history
- Environmental or occupational hazards

Headaches: COLDSPA
Dizziness/lightheadedness may be related to carotid artery disease, cerebellar abscess, Meniere disease, or inner ear infections
Do you fall with your lightheadedness? Do you experience syncope? Syncope is a temporary loss of consciousness. Imbalance and difficulty coordinating or controlling movements are seen in neurologic diseases involving the cerebellum, basal ganglia, or the vestibular part of cranial nerve 8 acoustic
Seizures: What happens before you have the seizure and where on your body the seizure starts…In some cases, an aura forewarns the client that a seizure is about to occur. Are they on antiepileptic medications? Are they therapeutic?
Tremors? The are some neurologic diseases with tremors…So are the tremors new? Where are they? Hands, legs? Is it worse with movement? Is it constant or does it subside with rest?
Weakness? Unilateral weakness or paralysis; did they have a CVA, compression of the spinal cord, or nerve injury. Progressive weakness is a symptom of several nervous system diseases.
Numbness or tingling: Use COLDSPA: loss of sensation, tingling, or burning may occur with damage to the brain, spinal cord, or peripheral nerves
Difficulty swallowing: May be related to a CVA, Parkinson's disease, myasthenia gravis, Guillain-Barre syndrome, or dysfunction of cranial nerves 9,10 or 12
Difficulty speaking: Injury to the cerebral cortex can impair the ability to speak or understand verbal language

Past History: Head injuries..think about football and the recurrent trauma to the brain.... Even minor injuries can produce long term neurologic deficits and affect the client's level of functioning
Prolonged exposure to lead, insecticides, pollutants or other chemicals can alter neurologic status

Cranial nerves



CN 1: Olfactory
CN 2: Optic
CN 3: Oculomotor
CN 4: Trochlear
CN 5: Trigeminal
CN 6: Abducens
CN 7: Facial
CN 8: Acoustic, vestibulocochlear
CN 9: Glossopharyngeal
CN 10: Vagus
CN 11: Spinal accessory
CN 12: hypoglossal

Cranial nerve I: Olfactory

- Occlude each nostril and test different smells
- Not done routinely
  - Test sense of smell in those who report:
  - Loss of smell with heat trauma
  - Abnormal mental functioning

Carries smell impulses from nasal mucous membrane to brain.
When you test the Olfactory nerve make sure the client's eyes are closed so that they can not see the object that they are smelling.
Inability to smell or identify the correct scent is an abnormal finding which can indicate a lesion or tumor in the frontal lobe. Some injury to the nerve from smoking or use of cocaine, chronic sinus infections or sinus problems

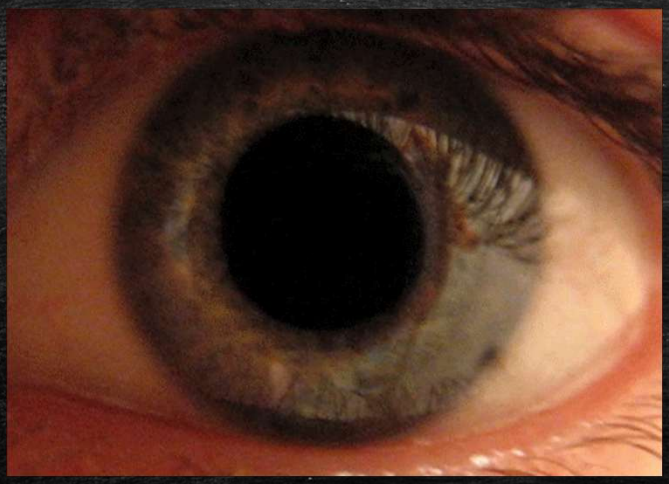
Cranial nerve II: Optic

- Test visual acuity with Snellen eye chart (distance)
  - Hand-held Jager chart (near)
- Screen visual fields by confrontation
- Use ophthalmoscope to examine ocular fundus to determine color, size, and shape of the optic disc

Carries visual impulses from eye to brain.
Test Snellen-distant Jager-near
Confrontation-visual fields
Using an ophthalmoscope to examine the optic disc: Round red reflex is present.

Abnormal findings of CNII can be seen with retinal damage, lesions of the optic nerve or the parietal cortex, or swelling of the optic nerve (papilledema) that occurs with increased intracranial pressure, intracranial hemorrhage, or a brain tumor.

Cranial nerves III, IV, & VI: Oculomotor, trochlear, & abducens



- Test extraocular movements in 6 cardinal fields of gaze
- Lid elevation
- Check convergence
- Assess pupillary response to light & accommodation

III (Oculomotor): Contracts eye muscles to control eye movements (interior lateral, medial, and superior), constricts pupils, and elevates eyelids
IV (Trochlear): Contracts one eye muscle to control inferomedial eye movement
VI (Abducens): Controls lateral eye movements

Lid elevation: Normal: eyelid covers about 2mm of the iris
Abnormal: ptosis…drooping of the eyelid which can be seen with Myasthenia gravis

Assess extraocular movements: Eye movements should be smooth, coordinated motion in all directions (six cardinal fields)
Abnormal: Nystagmus, seen with cerebellar disorders.
Limited eye movement seen with increased intracranial pressure. Paralysis of the oculomotor, trochlear, or abducens nerves.

Assess Pupillary response to light and accommodation in both eyes: Normal pupils constrict simultaneously.
Abnormal: Constricted, fixed pupils: Narcotic use or damage to the pons. Unilaterally dilates pupil unresponsive to light or accommodation: Damage to CNIII. Constricted pupil unresponsive to light or accommodation: Lesions of the sympathetic nervous system.

Cranial nerve V: Trigeminal

- Palpate temporal & masseter muscles while client clenches teeth
- Assess sharp/dull on forehead, cheek, and jaw
- Assess corneal reflex



Carries sensory impulses of pain, touch, and temperature from the face to the brain; influences clenching and lateral jaw movements (biting, chewing)
3 Parts to this test:
Test motor function. Ask the client to clench the teeth while you palpate the temporal and masseter muscles for contraction. Temporal and masseter muscles contract bilaterally.
Abnormal is decreased contraction in one or both sides. Asymmetric strength in moving the jaw may be seen with lesion or injury of the 5th cranial nerve. Pain occurs with clenching of the teeth. Bilateral muscle weakness is seen with peripheral or central nervous system dysfunction.
Test Sensory function: Touch the client's forehead, cheeks, and chin with a sharp or dull side of a paper clip. Normal findings: The client correctly identifies sharp and dull stimuli and light touch to the forehead, cheeks and chin.
Abnormal: Inability to feel and correctly identify facial stimuli occurs with lesions of the trigeminal nerve or lesions in the spinothalamic tract.
Test the corneal reflex: Ask the client to look away and up while you lightly touch the cornea with a fine wisp of cotton. Eyelids should blink bilaterally. An absent corneal reflex may be noted with lesions of the trigeminal nerve or lesions of the motor part of cranial nerve 7.

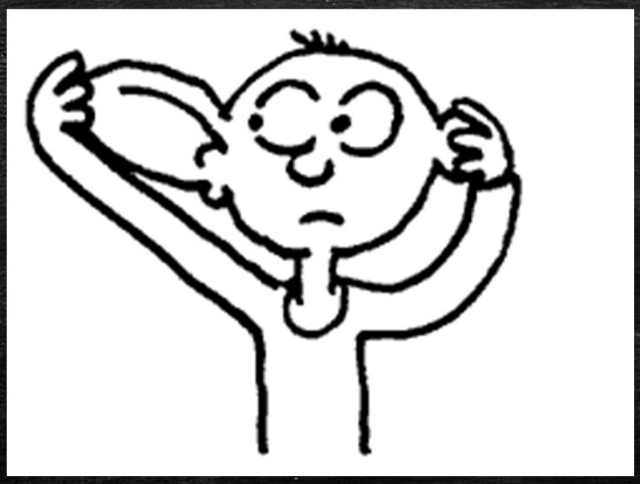
Cranial nerve VII: Facial

- Assess face for asymmetry, tics, or abnormal movements
- Ask client to:
  - Raise eyebrows
  - Frown
  - Close eyes tightly
  - Show teeth (grimace)
  - Smile
  - Puff both cheeks



Contains sensory fibers for taste on anterior two thirds of tongue and stimulates secretions from salivary glands (submaxillary and sublingual) and tears from lacrimal glands.
Supplies the facial muscles and affects facial expressions (smiling, frowning, closing eyes).
Assess motor function: by having the client smile, frown, wrinkle forehead, show teeth, puff out cheeks, purse lips, raise eyebrows, and close eyes tightly against resistance
Normal: All should be symmetric.
Abnormal: Inability to close eyes, wrinkle forehead and so forth. Can be seen with Bell Palsy or from the effects from a stroke.
Sensory function of CNVII not routinely tested.

Cranial nerve VIII: Acoustic



- Ability to hear
  - Normal conversation
  - Whispered voice
  - Weber & Rinne testing

Test the client's hearing ability in each ear & perform the Weber and Rinne tests to assess the cochlear component of CNVIII.

Should be able to hear whispered words from 1-2ft.
Do not worry about knowing the Weber & Rinne.

Cranial nerve IX & X: Glossopharyngeal & Vagus

- Assess:
  - If voice is hoarse
  - Swallowing
  - Movement of the soft palate & uvula as client says "Ah"
  - Gag reflex (warn client first)

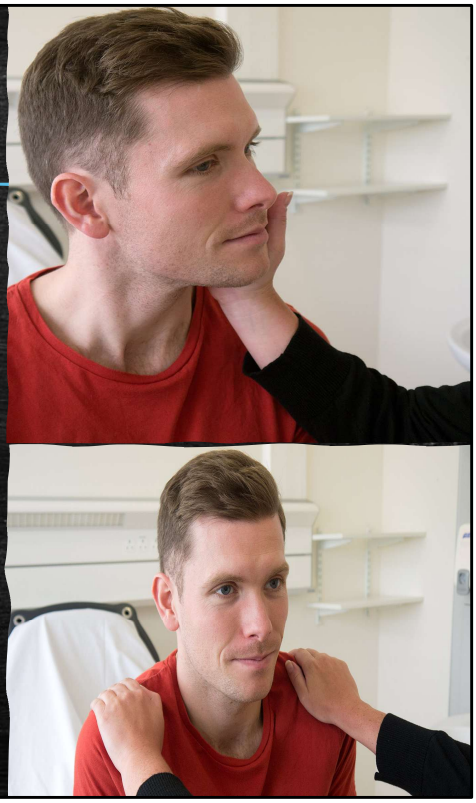


IX (Glossopharyngeal): Contains sensory fibers for taste on posterior third of tongue and sensory fibers of the pharynx that result in the gag reflex when stimulated. Provides secretory fibers to the parotid salivary glands; promotes swallowing movements

X (Vagus): Carries sensations from the throat, larynx, heart, lungs, bronchi, gastrointestinal tract, and abdominal viscera; promotes swallowing, talking, and production of digestive juices
CN: 9 and 10:
Test Motor Function: Ask the client to open mouth wide and say ah while you use a tongue depressor on the client's tongue. Normal: uvula and soft palate rise bilaterally and symmetrically
Abnormal: Soft palate does not rise with bilateral lesions of cranial nerve 10 vagus.
Test the Gag reflex: Touch the posterior pharynx with the tongue depressor.
Normal: gag reflex intact, some client may have a reduced or absent gag reflex. An absent reflex may be seen with lesions of cranial nerve 9 or 10.
Check ability to swallow: Client swallows without difficulty: Dysphagia or hoarseness may indicate a lesion of cranial nerve IX or X or other neurologic disorder.

Cranial nerve XI: Spinal accessory

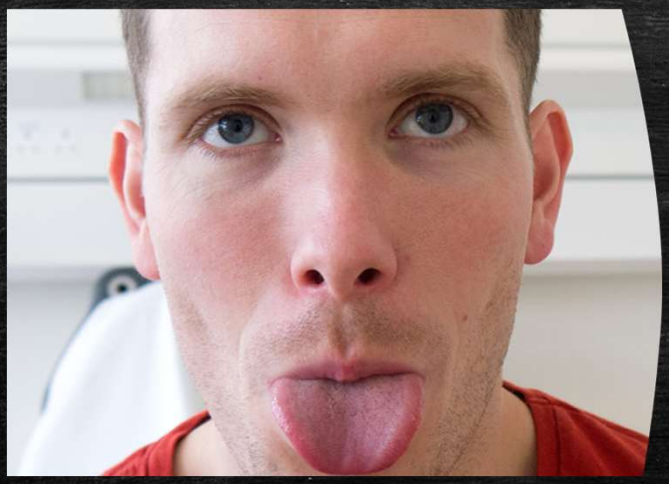
- Assess strength as client shrugs shoulders up against your hands. Note contractions of opposite sternocleidomastoid & force as client turns head against your hands.



Innervates neck muscles (sternocleidomastoid and trapezius) that promote movement of the shoulders and head rotation; also promotes some movement of the larynx.

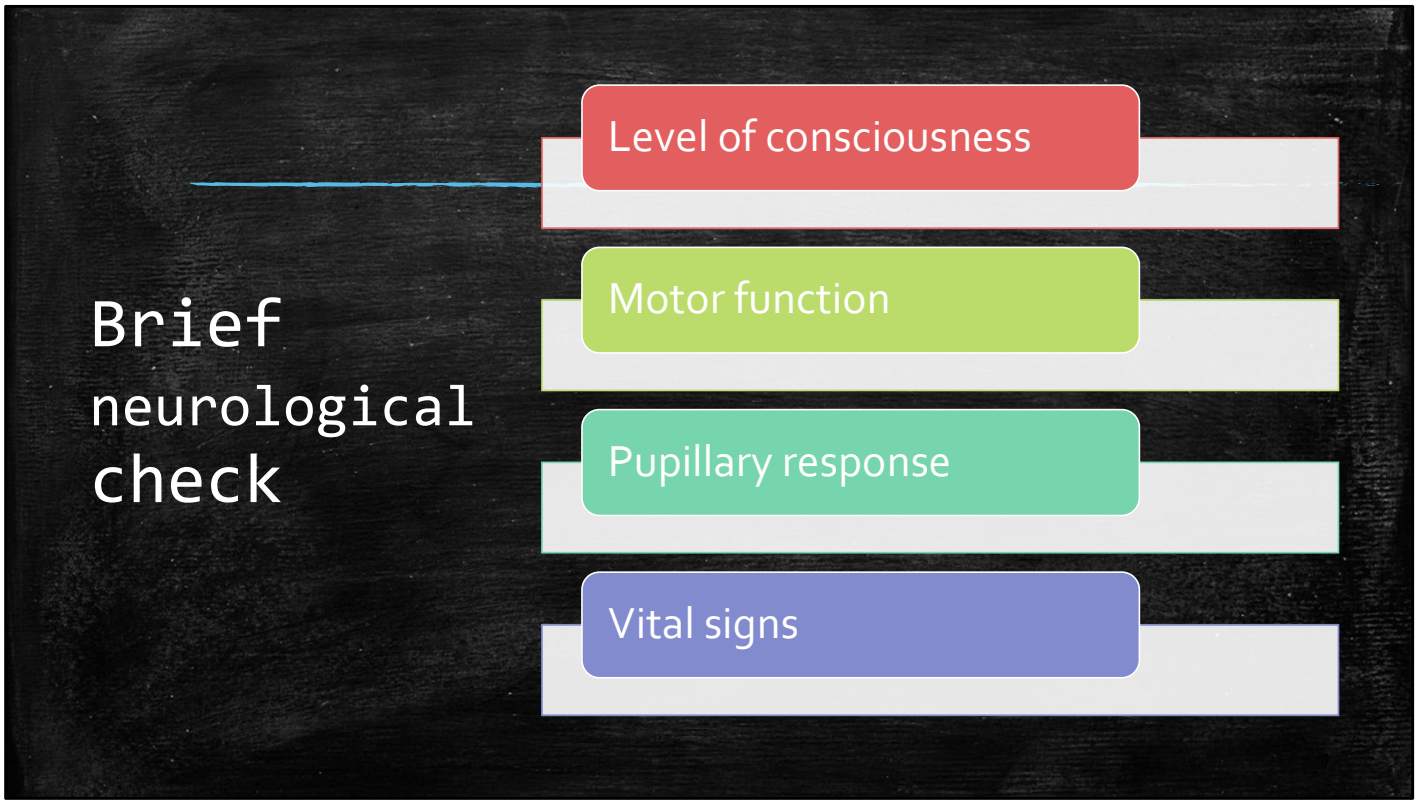
CN 11: Ask the client to shrug shoulders against resistance to assess the trapezius muscles: Normal: Symmetric, strong contraction of the trapezius muscles. Abnormal: asymmetric or drooping of the shoulder can be seen with paralysis or muscle weakness due to neck injury.
Ask the client to turn the head against resistance, first to the right and then to the left to assess the sternocleidomastoid muscles: Normal: strong contraction of the muscle. Abnormal: Atrophy may be seen with peripheral nerve disease.

Cranial nerve XII: Hypoglossal

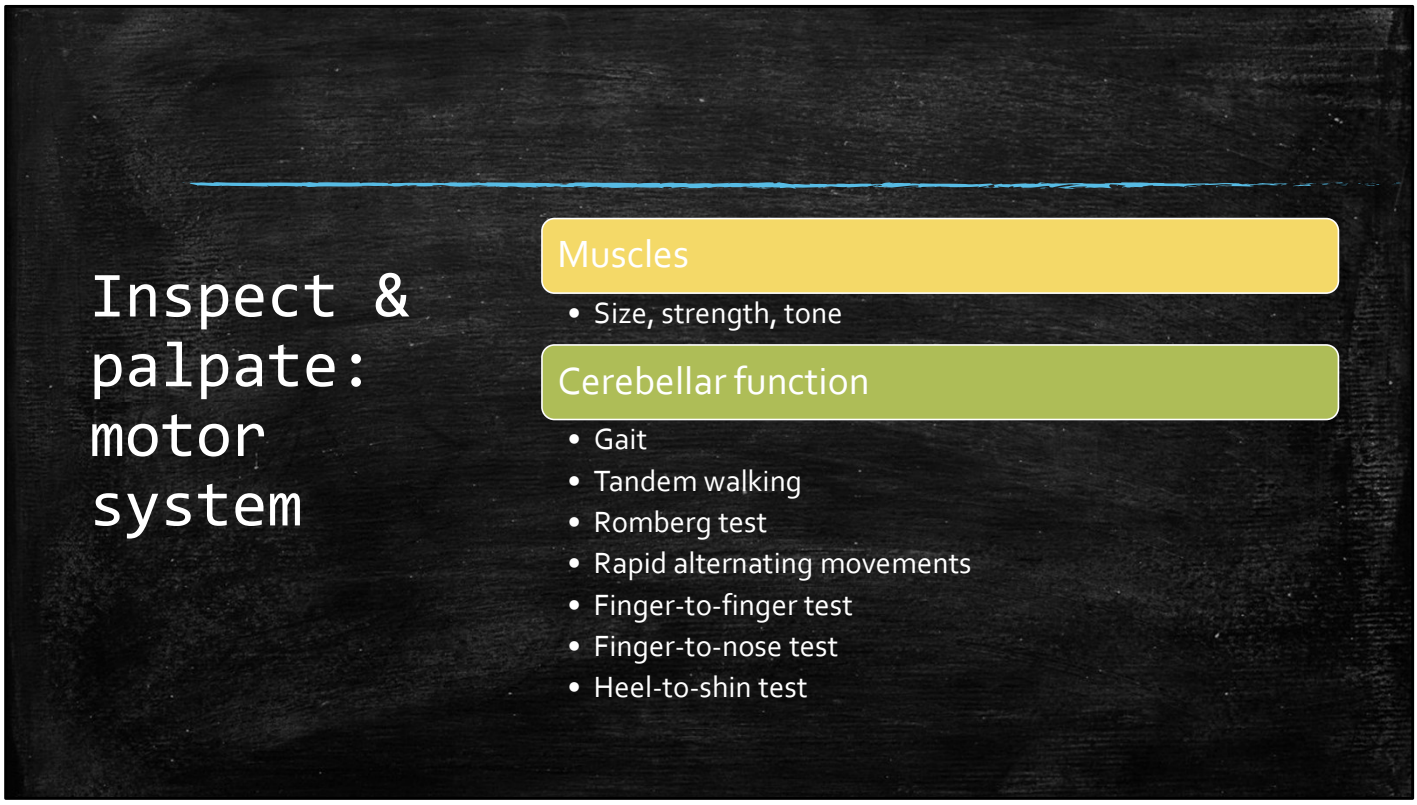


- Ask the client to protrude tongue & move it side-to-side
- Assess for symmetry & atrophy

Innervates tongue muscles that promote the movement of food and talking.
To assess the strength and mobility of the tongue, ask the client to protrude tongue, move it to each side against the resistance of a tongue depressor and then put it back in the mouth. Normal: symmetric and smooth and bilateral strength is apparent.
Abnormal: Fasciculations and atrophy of the tongue which would be seen with peripheral nerve damage. Deviation to the affected side is seen with a unilateral lesion.



Anytime you have a client come in with neurologic issues; you always want to recheck them or check them more often than a client who comes in for other reasons. Sometimes you are checking and rechecking their neuro status every 1, 2, or 4 hours. Those rechecks you are assessing the level of consciousness, motor function, pupillary response and vital signs



Motor and Cerebellar systems: assess the muscles size and symmetry. Are they fully developed, relaxed and contract voluntarily and show mild, smooth resistance to passive movement.
Involuntary movements is abnormal: tics, twitch, tremors
Motor: gait should be smooth and coordinated.. This is gross motor movement.
When we get to the finger-to-finger test…this is fine motor movement.

Muscle strength was covered in chapter 24

Inspection of gait

- Normal
  - Smooth, rhythmic, effortless, coordinated
- Tandem walking
  - Heel-to-toe
- Abnormal
  - Cerebellar ataxia
  - Scissors gait
  - Spastic hemiparesis (Hemiplegic)
  - Parkinsonian gait
  - Footdrop



Gait. Observe as the person walks 10 to 20 feet, turns, and returns to the starting point. Normally, the person moves with a sense of freedom. The gait is smooth, rhythmic, and effortless; the opposing arm swing is coordinated; the turns are smooth. The step length is about 15 inches from heel to heel.
Tandem walking may illicit an abnormal gait not seen with normal walking effort.
Cerebellar Ataxia: wide-based, staggering, unsteady gait, Romberg test results are positive, seen with cerebellar diseases or alcohol or drug intoxication
Scissor Gait: stiff, short gait, thighs overlap each other with each step, seen with partial paralysis of the legs
Spastic Hemiparesis (Hemiplegic): Flexed arm held close to body while client drags toe of leg or circles it stiffly outward and forward (circumduction of the foot). Seen with lesions of the upper motor neurons in the cortical spinal tract such as occurs in stroke.
Parkinsonian gait: Shuffling, turns stiff, stooped over posture with flexed hips & knees.
Footdrop: client lifts foot and knee high with each step, then slaps the foot down hard on the ground. Client cannot walk on heels, characteristic of diseases of the lower motor neurons.

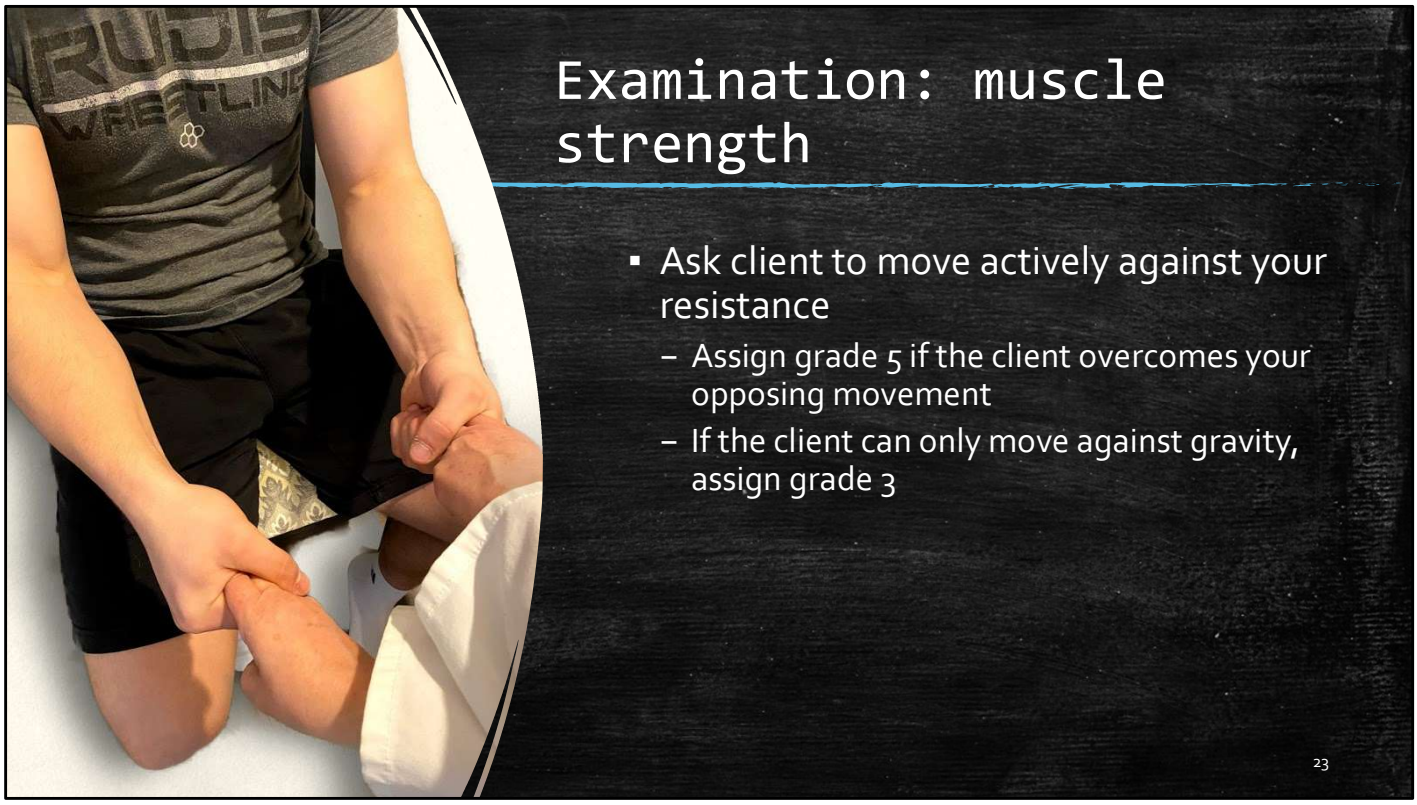
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Romberg test

- Ask client to stand with feet together and arms at sides
- Close eyes
- Normal: maintain posture & balance



The Romberg Test. Ask the person to stand up with feet together and arms at the sides. Note any unsteadiness or swaying. Then, ask the person to close the eyes and to hold the position. Wait about 20 seconds. Normally, a person can maintain posture and balance even with the visual orienting information blocked, although slight swaying may occur. (Stand close to catch the person in case they fall.)
Abnormal Findings: **Positive Romberg**
Sways, falls, widens base of feet to avoid falling.
Positive Romberg sign is loss of balance that occurs when closing the eyes. You eliminate the advantage of orientation with the eyes, which had compensated for sensory loss. A positive Romberg sign occurs with cerebellar ataxia (multiple sclerosis, alcohol intoxication), loss of proprioception, and loss of vestibular function.
Ask the person to perform a shallow knee bend or to hop in place, first on one leg, then the other. This demonstrates normal position sense, muscle strength, and cerebellar function. Note that some individuals cannot hop because of aging or obesity.



In our neurological assessment: Head to Toe Assessment we perform Hand grips and pedal push pulls.

We discussed grading muscle strength in chapter 24.

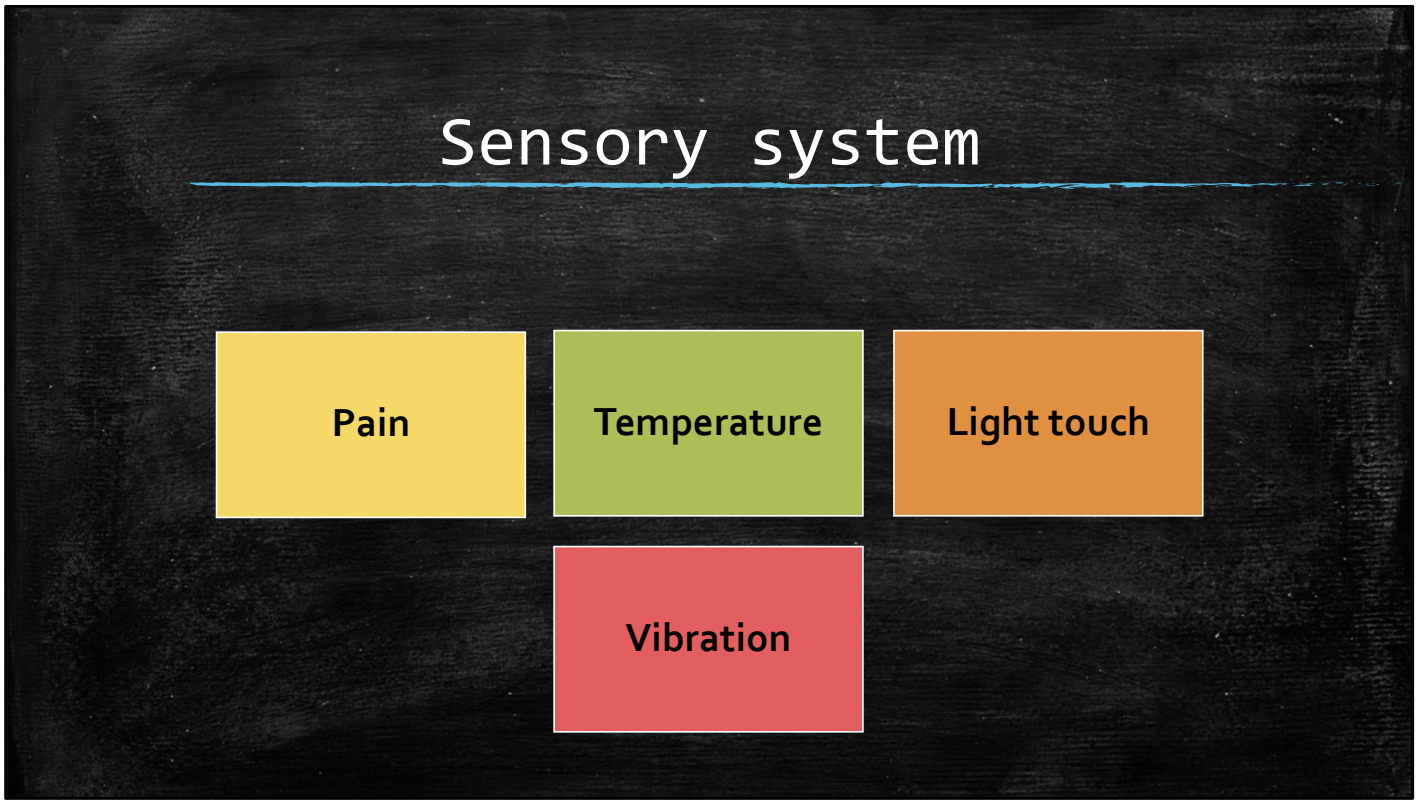
Rapid alternating movements/Finger-to-finger test
Finger to nose test/Heel to shin test

- Movements should be smooth and coordinated
- Inability to perform accurately can indicate dysdiadochokinesia (cerebellar disease)



Rapid Alternating Movements (RAM). Ask the person to pat the knees with both hands, lift up, turn hands over, and pat the knees with the backs of the hands. Then ask the person to do this faster. Normally, this is done with equal turning and a quick rhythmic pace.
Abnormal Findings
Lack of coordination.
Slow, clumsy, and sloppy response is termed dysdiadochokinesia and occurs with **cerebellar disease**.
Finger-to-finger test: ask the person to touch the thumb to each finger on the same hand, starting with the index finger, then reverse direction. Normally, this can be done quickly and accurately.
Abnormal Findings
Lack of coordination, tremors
Finger-to-Nose Test. Ask the person to close the eyes and to stretch out the arms. Ask the person to touch the tip of his or her nose with each index finger, alternating hands and increasing speed. Normally this is done with accurate and smooth movement.
Abnormal Findings: Misses nose. Worsening of coordination when the eyes are closed occurs with cerebellar disease.
Heel-to-Shin Test. Test lower extremity coordination by asking the person, who is in a supine position, to place the heel on the opposite knee, and run it down the shin from the knee to the ankle. Normally, the person moves the heel in a straight line

down the shin.
Abnormal Findings: Lack of coordination, Heel falls off shin.



Now that we have gone through the motor system; now we are going to explore the sensory system associated with the neurological assessment.

Many disorders can alter a person's ability to perceive sensations correctly. These include peripheral neuropathies (due to diabetes mellitus, folic acid deficiencies, and alcohol use disorder) and lesions of the ascending spinal cord, brain stem, cranial nerves, and cerebral cortex.

Assessing pain, light touch, & temperature



- Pain: Dull vs. Sharp
- Light touch: Cotton wisp to skin
- Temperature: Hot & cold
  - Only test if pain & light touch response is abnormal

Have client close eyes.
Pain: Dull vs Sharp. Use a paperclip or safety pin. Vary sites and touch.
Light Touch. Apply a wisp of cotton to the skin. Stretch a cotton ball to make a long end and brush it over the skin in a random order of sites and at irregular intervals. This prevents the person from responding just from repetition. Include the arms, forearms, hands, chest, thighs, and legs. Ask the person to say "now" or "yes" when touch is felt. Compare symmetric points.
Only test temperature if pain and light touch response is abnormal. Pain and temperature travel the same pathways, so if pain is intact, temperature will be also.

Vibratory sensation

- Use a tuning fork
- Begin distal
  - If no sensation, move proximal
- Compare right to left
- Often the first sensation lost



Vibration. Test the person's ability to feel vibrations of a tuning fork over bony prominences. Use a low-pitch tuning fork (128 Hz or 256 Hz) because its vibration has a slower decay. Strike the tuning fork on the heel of your hand and hold the base on a bony surface of the fingers and great toe. Ask the person to indicate when the vibration starts and stops. If the person feels the normal vibration or buzzing sensation on these distal areas, you may assume proximal spots are normal and proceed no further. If no vibrations are felt, move proximally and test ulnar processes, and ankles, patellae, and iliac crests. Compare the right side with the left side. If you find a deficit, note whether it is gradual or abrupt.
Abnormal Findings:
Unable to feel vibration. Loss of vibration sense occurs with peripheral neuropathy, e.g., **diabetes and alcoholism**. Often, this is the first sensation lost.
Peripheral neuropathy is worse at the feet and gradually improves as you move up the leg, as opposed to a specific nerve lesion, which has a clear zone of deficit for its dermatome.

Older adult considerations

- Aging adult
  - General atrophy with a steady loss of neurons in the brain and spinal cord
  - Neuron loss begins to appear after age 65
  - General loss of muscle tone
  - Diminished touch, pain sensation, taste & smell
- Impaired fine coordination & agility
- Loss of vibratory sense
- Decreased pupillary response
- Slowed reaction time
- General slowing of movement
- Decrease in vision & hearing

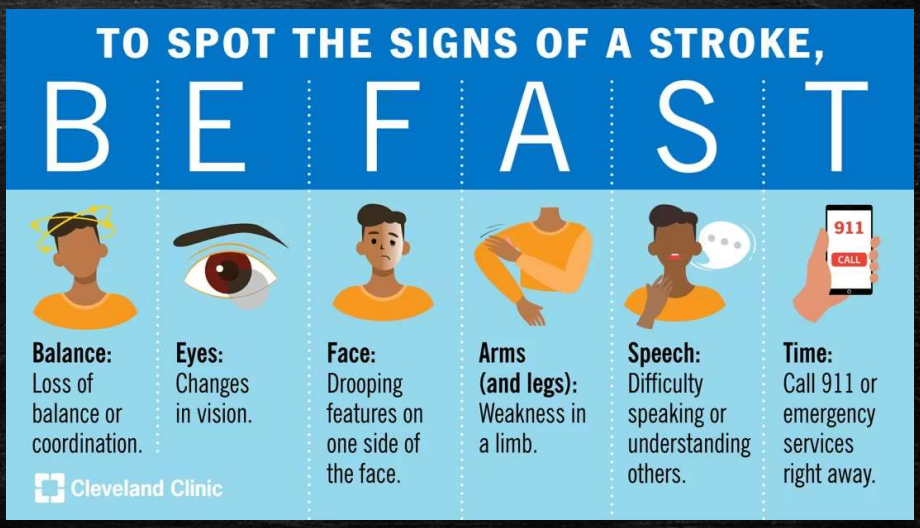
Health promotion and disease prevention: Cerebrovascular accident (CVA)

- Blood flow to a portion of the brain is interrupted or stops, depriving the brain cells of oxygen
- Types
  - Hemorrhagic
  - Ischemic
  - TIA (transient ischemic attack)
- Healthy People 2030 goal: Improve cardiovascular health and reduce deaths from heart disease and stroke

CVA: Stroke
Hemorrhagic strokes result when a brain aneurysm bursts or a weakened blood vessel in the brain leaks. Hemorrhagic strokes are less common (about 15% of strokes) but account for 40% of stroke deaths.
Ischemic strokes occur when a blood vessel carrying blood to the brain is blocked by a clot. These clots can be embolic (move through the vessel) or thrombotic (develop within the vessel).
TIA: A mini-stroke that causes no damage but indicates stroke risk is called a TIA. Stroke symptoms of a TIA usually last for a short time but may last up to 24 hours before symptoms disappear.

Cerebrovascular accident

- Symptoms



Cerebrovascular accident

- Risk factors: Modifiable
  - HTN, obesity, sedentary lifestyle, poor diet, smoking, DM, high cholesterol
- Risk factors: Non-modifiable
  - Age, family history/genetics/AB blood type, anxiety/depression, sleep apnea, prolonged exposure to high levels of air pollution, race, ethnicity, sex
- Client education:
  - Stop smoking, control high cholesterol/HTN/DM, exercise at least 30 minutes per day, maintain healthy weight, eat a healthy diet, limit alcohol

Diet recommendations:
Choose a diet rich in fruits, vegetables, and whole grains.
Choose lean proteins, such as chicken, fish, beans, and legumes.
Choose low-fat dairy products, such as 1% milk and other low-fat items.
Avoid sodium (salt) and fats found in fried foods, processed foods, and baked goods.
Eat fewer animal products and foods that contain cheese, cream, or eggs.
Read labels and stay away from saturated fat and anything that contains partially hydrogenated or hydrogenated fats. These products are usually loaded with unhealthy fats.